Manō: Privacy-Preserving Synthetic Mental Health Data Generation and Intervention Simulation System

Project ID: 25-26J-163

Project Proposal Report

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Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

28.08.2025

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Signature of the co-supervisor: Date:

ABSTRACT

The field of mental health research is often hampered by the lack of large-scale, diverse, and accessible datasets due to stringent privacy regulations and the sensitive nature of the data. This limitation significantly hinders the development and validation of advanced computational models for understanding, diagnosing, and treating mental health conditions. This proposal introduces the Privacy-Preserving Mental Health Simulation System, a core component of the "Manō: Holistic Framework for Personalized and Community-Driven Cognitive, Emotional, and Resilient Vitality." Addressing critical needs in accessible and privacy-aware mental health support, this system generates high-fidelity, privacy-guaranteed synthetic mental health data and simulates intervention outcomes. It aims to advance research, model training, and intervention prototyping within the Manō ecosystem.

The system utilizes a hybrid Conditional Generative Adversarial Network (CGAN) architecture, proficient in handling multimodal data. This includes specialized sub-generators: CTGAN for tabular data, TimeGAN for time-series data, and Text-GANs for emotional texts. A unified discriminator evaluates the combined synthetic output, providing crucial feedback for model refinement. Integral to the system is the robust integration of differential privacy. Differentially Private Stochastic Gradient Descent (DP-SGD) is applied during GAN training, employing gradient clipping and noise addition to ensure strong, quantifiable privacy guarantees (ε, δ). The system also incorporates advanced privacy refinements and rigorous validation metrics, such as statistical similarity (MMD), machine learning utility, and privacy metrics (k-anonymity, membership inference attacks), to ensure both data integrity and privacy. Furthermore, an advanced Intervention Simulation Module models intervention effects on synthetic user profiles. This module employs Reinforcement Learning (RL) for optimal strategy learning, Causal Inference for discerning causality, and Agent-Based Modeling (ABM) for community-level simulations. A continuous feedback loop, leveraging real-world interaction data from Manō components like the chatbot, refines GAN simulations, thereby enhancing realism and applicability. This comprehensive approach ensures ethical data utilization and significantly contributes to personalized and community-driven mental wellness support.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ABM	Agent-Based Modeling
ADS-GAN	Anonymization through Data Synthesis using Generative Adversarial Networks
API	Application Programming Interface
CGAN	Conditional Generative Adversarial Network
CTGAN	Conditional Tabular GAN
DASS	Depression Anxiety Stress Scales
DP	Differential Privacy
DP-SGD	Differentially Private Stochastic Gradient Descent
DTx	Digital Therapeutics
EEG	Electroencephalogram
EHR	Electronic Health Records
GAN	Generative Adversarial Network
GMM	Gaussian Mixture Model
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
MIA	Membership Inference Attacks
MMD	Maximum Mean Discrepancy
NLP	Natural Language Processing
PaaS	Platform-as-a-Service
PATE	Private Aggregation of Teacher Ensembles
PHQ-9	Patient Health Questionnaire-9
RL	Reinforcement Learning
RNN	Recurrent Neural Network
SaaS	Software-as-a-Service
Text-GAN	Text Generative Adversarial Network
TimeGAN	Time-series Generative Adversarial Network
V-GAN	Variational Generative Adversarial Network

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1. INTRODUCTION

Mental health represents a significant global challenge, with approximately 13% of the world's population affected by mental health and substance use disorders [1]. The economic impact is substantial, surpassing 2.5 trillion annually and projected to reach 6 trillion by 2030 [1]. Despite this, governments allocate only about 2% of health budgets to mental health, and access to care remains limited, particularly in low- and middle-income regions [1]. This highlights a critical gap in accessible, scalable, and privacy-preserving mental health support systems. Existing digital mental health solutions often face challenges related to data privacy, lack of personalization, and limited integration of diverse support mechanisms. The sensitive nature of mental health data poses significant privacy concerns, making it difficult to collect and utilize real-world datasets for research and model development [2], [3]. While synthetic data generation using Generative Adversarial Networks (GANs) offers a promising avenue to overcome data scarcity and privacy issues, ensuring the fidelity and utility of such synthetic data while maintaining robust privacy guarantees remains a research challenge [4], [5].

This project, "Manō: Holistic Framework for Personalized and Community-Driven Cognitive, Emotional, and Resilient Vitality," aims to address these multifaceted challenges by developing a holistic framework that leverages cutting-edge machine learning techniques to deliver personalized, privacy-preserving, and community-driven cognitive and emotional health support. My component, the Privacy-Preserving Mental Health Simulation System, is a pivotal part of this framework. It is designed to generate high-fidelity, privacy-preserving synthetic mental health data and simulate the outcomes of various mental health interventions. This synthetic data will then be utilized by other Manō components for research, model training, and intervention prototyping. To achieve this, the project necessitates mastering several relevant studies and techniques. The core of this component lies in the development of a sophisticated Conditional GAN (CGAN) architecture capable of generating diverse synthetic mental health data, including survey responses, wearable metrics, and emotional texts [6]. The

state of the art in this domain involves hybrid GAN architectures that can handle multimodal data, such as CTGAN for tabular data, Time GAN for time-series data, and specialized Text-GANs for textual data [6]. Furthermore, integrating differential privacy mechanisms, such as Differentially Private Stochastic Gradient Descent (DP-SGD), is crucial to ensure privacy guarantees for the synthetic data, building upon existing work in privacy-preserving AI [7, 8]. The approach will also incorporate advanced intervention simulation models, including Reinforcement Learning and Causal Inference, to model the effects of mental health interventions, extending beyond simple rule-based simulations [9, 10]. This project builds upon previous work by integrating these advanced techniques into a unified system, focusing on both data utility and strong privacy guarantees, which is a critical advancement in the field of digital mental health.

1.1 Background & Literature Survey

Mental health is a complex and multifaceted domain, increasingly recognized as a critical public health concern. The global burden of mental health disorders is substantial, impacting individuals, communities, and economies [1]. Digital mental health solutions have emerged as a promising avenue to address accessibility and scalability challenges in traditional care [2]. However, the sensitive nature of mental health data necessitates robust privacy-preserving mechanisms to foster trust and encourage adoption [3]..

Generative adversarial networks (GANs) for synthetic data generation

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. [4], have revolutionized synthetic data generation. A GAN consists of two neural networks: a generator that creates synthetic data and a discriminator that distinguishes between real and synthetic data. Through an adversarial training process, the generator learns to produce data that is indistinguishable from real data. In the context of mental health, GANs offer a powerful tool to overcome data scarcity and privacy concerns by generating high-fidelity synthetic datasets that mimic the statistical properties of real patient data without exposing sensitive information [4], [5]. For multimodal mental

health data, which includes structured (e.g., survey responses), time-series (e.g., wearable sensor data), and unstructured text data (e.g., emotional texts), specialized GAN architectures are required. Conditional GANs (CGANs) allow for the generation of data conditioned on specific attributes, enabling the creation of synthetic profiles for particular user demographics or mental health conditions [6]. Specifically, Conditional Tabular GAN (CTGAN) has shown efficacy in generating high quality synthetic tabular data, preserving complex relationships within structured datasets [6]. For time-series data, TimeGAN is a prominent architecture capable of capturing temporal dynamics and generating realistic sequences, crucial for wearable sensor data in mental health applications [6]. Generating coherent and contextually relevant emotional texts is challenging, and specialized Text-GANs or Variational GANs (V-GANs) are employed, often integrating recurrent neural networks (RNNs) or transformer architectures to capture linguistic and semantic properties [6]. The integration of these specialized GANs into a hybrid architecture is a cutting-edge approach to handle the inherent complexity and diversity of mental health data [1].

Differential privacy in synthetic data

While GANs can generate realistic synthetic data, ensuring that this data does not inadvertently leak sensitive information about the training set is paramount. Differential Privacy (DP) provides a rigorous, mathematical framework for quantifying and guaranteeing privacy in data analysis and machine learning [2]. The core principle of DP is to add carefully calibrated noise to data or computations, making it statistically impossible to infer whether any single individual's data was included in the dataset or contributed to a model's output. This provides a strong privacy guarantee, protecting against various privacy attacks, including re-identification and membership inference [3], [4].

Differentially Private Stochastic Gradient Descent (DP-SGD) is a widely adopted method for achieving differential privacy in deep learning models, including GANs [5]. DP-SGD involves clipping gradients to limit the influence of individual data points and adding Gaussian noise during the training process. The privacy guarantee

is quantified by epsilon (ε) and delta (δ) parameters, where a smaller ε indicates stronger privacy [10]. Beyond DP-SGD, other mechanisms like Private Aggregation of Teacher Ensembles (PATE) have been explored to offer stronger privacy guarantees in certain scenarios by training multiple teacher models on disjoint subsets of sensitive data and aggregating their noisy predictions [7]. The rigorous evaluation of DP mechanisms involves not only calculating privacy budgets (ε, δ) but also assessing the utility of the synthetic data through statistical similarity metrics, machine learning utility, and vulnerability to privacy attacks like membership inference [8], [9].

Intervention simulation and personalized mental health

Beyond static data generation, simulating the effects of mental health interventions is crucial for developing personalized and effective support systems. Traditional rule-based simulations often lack the nuance and adaptability required for complex mental health trajectories. Advanced data-driven models, such as Reinforcement Learning (RL), offer a powerful approach to model adaptive interventions [10]. In an RL framework, an agent can learn optimal intervention strategies by interacting with a synthetic environment, where the 'state' represents a user's mental health profile, 'actions' are different interventions, and 'rewards' are defined by positive changes in mental health metrics [1]. This allows for the dynamic prediction of intervention outcomes and the generation of synthetic data that reflects these changes.

Causal inference models are also vital for understanding the true impact of interventions, helping to distinguish correlation from causation [2]. Techniques from causal inference can analyze relationships between intervention types, user characteristics, and outcomes in synthetic data to infer causal links, providing more robust insights into intervention effectiveness. Furthermore, Agent-Based Modeling (ABM) can be employed to simulate the effects of interventions within a community context, allowing for the study of how individual interventions propagate through a synthetic community and affect overall mental well-being and resilience [3]. These advanced simulation techniques move beyond simple predictive models, enabling a

more comprehensive and realistic understanding of mental health dynamics and intervention efficacy.

1.2 Research Gap

1.2.1 Table related to research

Characteristic	Data type	Generative Model	Privacy Technique	Intervention Simulation	Mental Health Focus
Yoon et al. [4]	Tabular EHR data	ADS-GAN	GAN-based anonymization	No	No
Sun et al. [3]	Medical data	Not specified	Differential Privacy	No	No
Akpinar et al. [1]	Medical images/signals	GANs	Not primary focus	No	No
Al-Itejawi et al. [2]	EEG signals	GANs	Not primary focus	No	No
Proposed System	Multimodal Mental Health Data	Conditional GAN (CGAN)	Differential Privacy (DP-SGD)	Yes	Yes

Table 1.2.1: Characteristics of existing research literature and proposed system

Despite significant advancements in digital mental health solutions and synthetic data generation, several critical gaps persist that this project aims to address:
Limited Integration of Multimodal Data with Robust Privacy Guarantees: While GANs can generate synthetic data, and differential privacy mechanisms exist, there is a research gap in comprehensively integrating these for multimodal mental health data (tabular, time-series, and text) while ensuring strong, quantifiable privacy guarantees and maintaining high data utility. Existing solutions often compromise on one aspect for the sake of another, or do not fully address the complexity of multimodal data in a privacy-preserving manner.

1. **Lack of Dynamic Intervention Simulation and Feedback Loops:** Current synthetic data generation often focuses on static data. There is a gap in dynamic

simulation systems that can accurately model the effects of various mental health interventions on synthetic user profiles and, crucially, incorporate a continuous feedback loop from real-world interactions (e.g., from a chatbot) to refine these simulations. This feedback mechanism is vital for improving the realism and applicability of synthetic data for intervention prototyping and research.

2. **Limited Integration of Multimodal Data with Robust Privacy Guarantees:** While GANs can generate synthetic data, and differential privacy mechanisms exist, there is a research gap in comprehensively integrating these for multimodal mental health data (tabular, time-series, and text) while ensuring strong, quantifiable privacy guarantees and maintaining high data utility. Existing solutions often compromise on one aspect for the sake of another, or do not fully address the complexity of multimodal data in a privacy-preserving manner.

3. **Insufficient Focus on Causal Inference in Synthetic Environments:** Many predictive models identify correlations but fall short in establishing causal relationships between interventions and outcomes. A significant gap exists in integrating causal inference models within synthetic data generation frameworks to understand the true impact of interventions, allowing for more evidence- based and personalized mental health support.

4. **Limited Community-Level Simulation and Resilience Modeling:** While individual-level simulations are progressing, there is a need for models that can simulate the propagation of interventions and their effects within a community context. The current landscape lacks robust Agent-Based Models integrated with privacy-preserving synthetic data generation to understand and foster community-driven resilience.

This project directly addresses these gaps by developing a Privacy-Preserving Mental Health Simulation System that not only generates high-fidelity, multimodal synthetic data with strong differential privacy guarantees but also dynamically simulates intervention outcomes with a continuous feedback loop and incorporates causal

inference and community-level modeling capabilities. This holistic approach aims to bridge the divide between privacy, data utility, and the dynamic nature of mental health interventions.

1.3 Research Problem

The overarching research problem addressed by the "Manō" project is the critical need for accessible, scalable, and privacy-preserving mental health support systems in the face of a growing global mental health crisis. Within this broader context, the specific research problem that the Privacy-Preserving Mental Health Simulation System component aims to solve can be articulated as follows:

How can high-fidelity, multimodal synthetic mental health data be generated and utilized to simulate the outcomes of various mental health interventions, while rigorously preserving individual privacy and ensuring the statistical utility and ethical soundness of the generated data for research, model training, and intervention prototyping?

This problem is compounded by several challenges:

Data Scarcity and Sensitivity: Real-world mental health data is inherently sensitive, making its collection, sharing, and utilization for research and model development highly challenging due to privacy regulations and ethical concerns. This leads to a scarcity of comprehensive datasets for training robust AI models.

Multimodality and Complexity: Mental health data is diverse, encompassing structured survey responses, time-series physiological signals from wearables, and unstructured textual expressions. Generating synthetic data that accurately captures the complex interdependencies and temporal dynamics across these modalities is a significant technical hurdle.

Privacy-Utility Trade-off: Implementing strong privacy guarantees, such as differential privacy, often comes at the cost of data utility. A key challenge is to develop mechanisms that provide robust privacy without significantly degrading the statistical properties and usefulness of the synthetic data for downstream analytical tasks and machine learning model training.

Dynamic Intervention Modeling: Simulating the dynamic effects of mental health interventions requires models that can predict changes in a user's mental state over time in response to different interventions. This goes beyond static data generation and demands sophisticated modeling techniques that can capture causal relationships and adaptive responses.

Ethical Deployment and Trust: For synthetic data to be widely adopted in mental health research and application development, there must be a high degree of trust in its privacy-preserving properties and its ability to accurately reflect real-world scenarios without introducing bias or compromising patient confidentiality.

This component directly confronts these challenges by developing a system capable of generating realistic, privacy-preserving synthetic mental health data that can be used to simulate and analyze the impact of various interventions, thereby providing a safe and ethical alternative to real data for advancing mental health research and personalized support.

2. OBJECTIVES

2.1 Main Objectives

The primary objective of the Privacy-Preserving Mental Health Simulation System component is to develop a robust, ethical, and high-fidelity system capable of generating privacy-preserving synthetic mental health data and simulating the outcomes of various mental health interventions, thereby facilitating research, model training, and intervention prototyping within the broader Manō framework.

2.2 Specific Objectives

To achieve the main objective, the following specific objectives will be pursued:

SO1: Design and Implement a Conditional GAN Architecture for Multimodal Data Generation: To develop and implement a sophisticated Conditional Generative Adversarial Network (CGAN) capable of generating diverse synthetic mental health data, including structured survey responses, time-series wearable metrics, and unstructured emotional texts, ensuring internal consistency across modalities.

SO2: Integrate Differential Privacy Mechanisms into the GAN Training Process: To incorporate robust differential privacy mechanisms, such as Differentially Private Stochastic Gradient Descent (DP-SGD) or Private Aggregation of Teacher Ensembles (PATE), into the CGAN training process to ensure strong, quantifiable privacy guarantees for the generated synthetic data.

SO3: Develop a Module to Simulate the Effects of Specific Mental Health Interventions: To create an advanced intervention simulation module that can accurately model and predict the effects of various mental health interventions on synthetic user profiles, utilizing techniques such as Reinforcement Learning and Causal Inference.

SO4: Establish a Feedback Loop Mechanism to Refine GAN Simulations: To implement a continuous feedback loop mechanism that refines the GAN simulations based on real-world interaction data from other Manō components (e.g., the chatbot), thereby enhancing the realism and applicability of the synthetic data.

SO5: Validate the Statistical Similarity and Utility of the Generated Synthetic Data: To rigorously validate the generated synthetic data for statistical similarity to real data and its utility for downstream tasks, including machine learning model training and research, while also assessing privacy risks through metrics like k-anonymity and membership inference attacks.

3. METHODOLOGY

The Privacy-Preserving Mental Health Simulation System, as a core component of the Manō project, will be developed using a systematic methodology encompassing data acquisition, model design and implementation, privacy integration, intervention simulation, and rigorous validation. The overall system architecture is designed to be modular and interconnected with other Manō components.

3.1 Overall System Description

The Privacy-Preserving Mental Health Simulation System is centered around a sophisticated Conditional Generative Adversarial Network (CGAN) designed to handle multimodal mental health data. This system will generate high-fidelity, privacy- preserving synthetic data that accurately reflects real-world mental health profiles and simulates the effects of various interventions. The generated synthetic data will serve as a crucial resource for training other Manō components, such as the Stress and Cognitive Risk Prediction System (LSTM), the Empathetic Conversational Support System (NLP Chatbot), and the Community-Driven Resilience Clustering System (GMM).

3.2 System Components and Interactions

1. **Multimodal Data Ingestion and Preprocessing:** Raw mental health data from various sources (e.g., DASS responses, Mental Health in Tech surveys, Sleep Health and Lifestyle data, Wearable IoT Sensors, Reddit depression posts) will be ingested and preprocessed. This involves handling missing values, normalizing data, harmonizing categorical variables, and extracting relevant features to create a unified feature space. This step is crucial for preparing the diverse datasets for GAN training.

2. **Hybrid Conditional GAN (CGAN) Architecture:** The core of the system is a hybrid CGAN, comprising specialized sub-generators for different data modalities:

CTGAN-like Module: For generating synthetic tabular data, such as survey responses (e.g., PHQ-9 scores, demographic information).

Time GAN-like Module: For generating synthetic time-series data, such as wearable metrics (e.g., heart rate variability, sleep patterns).

Text-GAN-like Module: For generating synthetic unstructured text data, such as emotional texts (e.g., journal entries, chatbot interactions). These sub-generators will receive a common conditional input (e.g., a synthetic user profile, intervention parameters) to guide data generation. A unified discriminator will evaluate the realism and internal consistency of the combined synthetic data across all modalities.

3. **Differential Privacy (DP) Integration**: To ensure privacy guarantees, differential privacy mechanisms will be integrated into the GAN training process. Primarily, Differentially Private Stochastic Gradient Descent (DP-SGD) will be applied to both the generator and discriminator networks. This involves gradient clipping and the addition of calibrated noise to gradients, ensuring that the synthetic data does not inadvertently reveal sensitive information about the real individuals whose data was used for training. The privacy budget (ε, δ) will be carefully managed and quantified.

4. **Intervention Simulation Module:** This module will simulate the effects of specific mental health interventions on synthetic user profiles. It will leverage advanced models:

Reinforcement Learning (RL): An RL agent will learn optimal intervention strategies by interacting with the synthetic environment, where the synthetic user's mental health profile is the state, interventions are actions, and positive changes in mental health metrics are rewards. This allows the GAN to generate post-intervention data reflecting expected changes.

Causal Inference Models: These models will be used to distinguish correlation from causation, analyzing relationships between intervention types, user characteristics, and outcomes in the synthetic data to infer causal links.

Agent-Based Modeling (ABM): For community-level simulations, ABM will model how interventions propagate through a synthetic community, affecting overall mental well-being and resilience.

5. **Feedback Loop Mechanism:** A continuous feedback loop will be established with other Manō components, particularly the Empathetic Conversational Support System (chatbot). Real-world interaction data from the chatbot will be used to refine the GAN simulations, improving the realism and applicability of the generated synthetic data for intervention prototyping and research.

6. **Validation Framework:** The generated synthetic data will undergo rigorous validation. This includes:
Statistical Similarity: Comparing statistical properties (mean, variance, correlations, distributions) of synthetic data to real data using metrics like Maximum Mean Discrepancy (MMD).

Machine Learning Utility: Training downstream ML models (e.g., Stress and Cognitive Risk Prediction, Resilience Clustering) in synthetic data and comparing their performance to models trained on real data.

Privacy Metrics: Evaluating privacy risks using k-anonymity, l-diversity, t- closeness, and conducting simulated Membership Inference Attacks (MIA) and Re-identification Attacks.

3.3 System Diagram

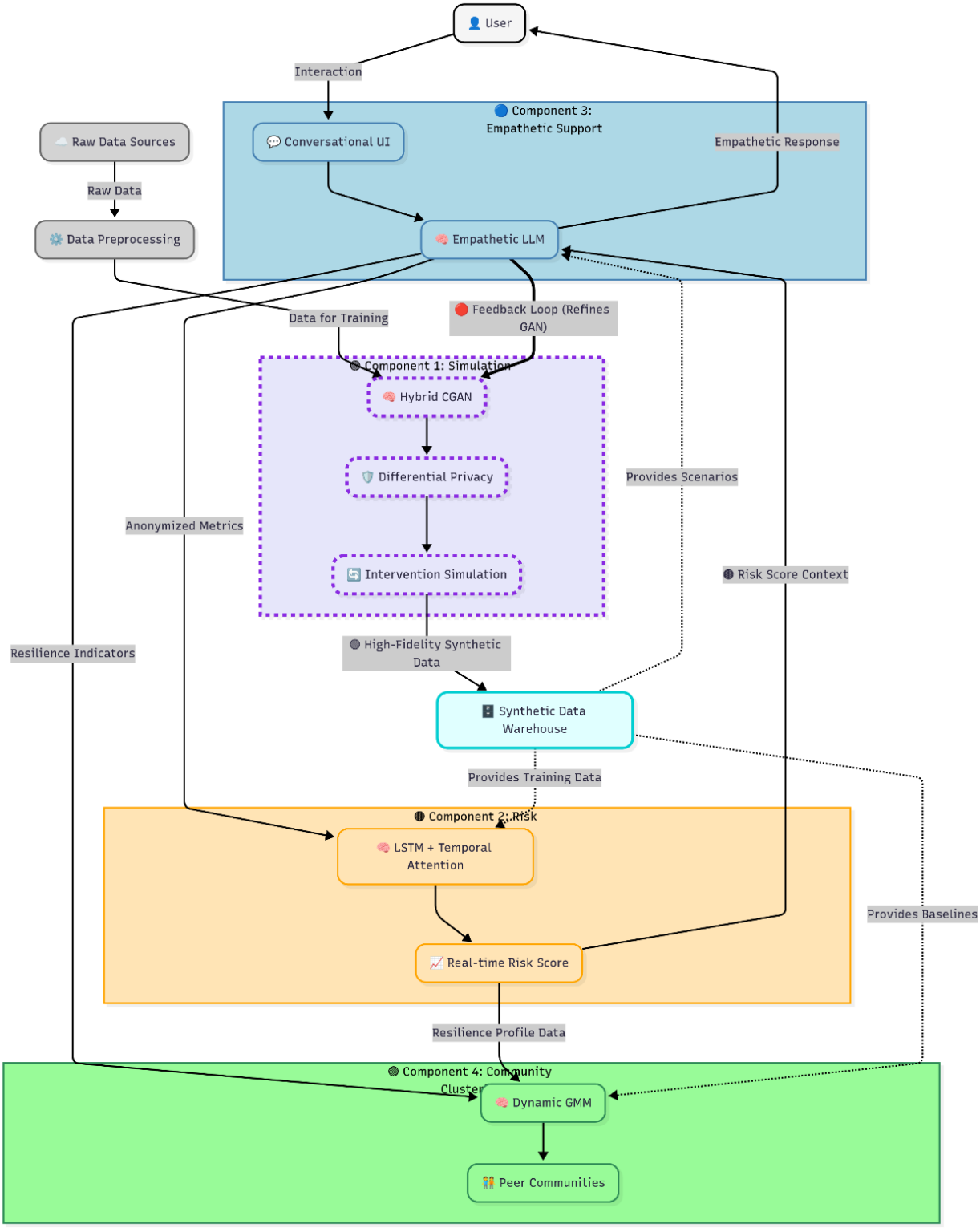

Figure 3.3.1: System overview diagram

3.4 Individual System Diagram

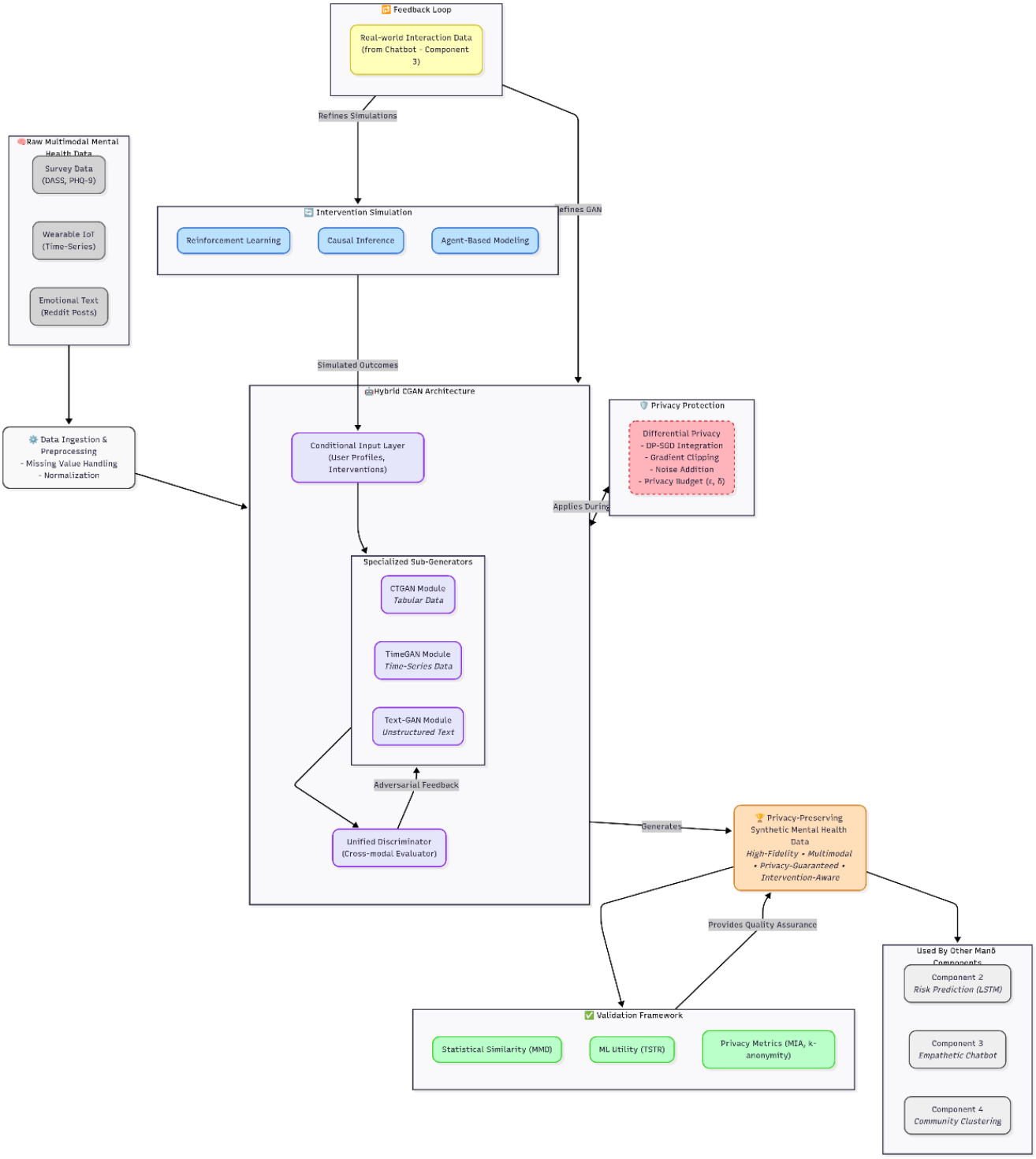

Figure 3.4.1: Individual system diagram

3.5 Explanation of The Individual System Diagram

The diagram illustrates the flow of data and the interaction between different modules within the Privacy-Preserving Mental Health Simulation System and its connection to other Manō components. Raw multimodal data is preprocessed and fed into the Hybrid CGAN. The CGAN, with its specialized sub-generators, produces synthetic data across different modalities. The Unified Discriminator provides adversarial feedback to the CGAN, improving data realism. The Intervention Simulation Module, driven by RL, Causal Inference, and ABM, models intervention outcomes, which also feed back into the CGAN for dynamic data generation. The privacy-preserving synthetic data generated by the system is then utilized by other Manō components (LSTM, Chatbot, GMM). Crucially, real-world interaction data from the Chatbot provides a continuous feedback loop to both the Intervention Simulation Module and the CGAN, enabling adaptive refinement. Finally, a comprehensive Validation Framework ensures the quality, utility, and privacy guarantees of the synthetic data, informing further refinements.

4. PROJECT REQUIREMENTS

This section outlines the key requirements for the Privacy-Preserving Mental Health Simulation System, categorized into functional, non-functional, and expected test cases.

4.1 Functional Requirements

Functional requirements define what the system must do to achieve its objectives.

FR1: Multimodal Synthetic Data Generation: The system shall generate synthetic mental health data across multiple modalities, including structured tabular data (e.g., survey responses, demographic information), time-series data (e.g., wearable sensor metrics like heart rate variability, sleep patterns), and unstructured text data (e.g., emotional texts, journal entries).

FR2: Conditional Data Generation: The system shall allow for the generation of synthetic data conditioned on specific user profiles, mental health conditions, or intervention parameters.

FR3: Differential Privacy Integration: The system shall integrate differential privacy mechanisms (e.g., DP-SGD) during the GAN training process to ensure that the generated synthetic data provides strong, quantifiable privacy guarantees, preventing re-identification and membership inference attacks.

FR4: Intervention Outcome Simul6tion: The system shall simulate the effects of predefined mental health interventions on synthetic user profiles, generating post-

intervention synthetic data that reflects expected changes in mental health metrics.

FR5: Feedback Loop Integration: The system shall incorporate a mechanism to receive real-world interaction data (e.g., from the Manō chatbot) to continuously refine and

improve the realism and applicability of the GAN-generated synthetic data and intervention simulations.

FR6: Data Utility Preservation: The system shall ensure that the generated synthetic data maintains high statistical similarity to real data and is useful for downstream tasks, such as training machine learning models for stress prediction, resilience clustering, and conversational AI.

FR7: Data Export Capability: The system shall provide functionality to export the generated synthetic data in commonly used formats (e.g., CSV, JSON) for use by other Manō components or external researchers.

4.2 Non-Functional Requirements

Non-functional requirements specify criteria that can be used to judge the operation of a system, rather than specific behaviors.

NFR1: Performance: The system shall generate synthetic data for a given batch size within a reasonable timeframe (e.g., synthetic data generation for 1000 records within 5 minutes).

NFR2: Scalability: The system shall be capable of scaling to generate large volumes of synthetic data (e.g., millions of records) and handle an increasing number of data modalities and intervention types without significant degradation in performance.

NFR3: Privacy Assurance: The system shall provide a quantifiable privacy guarantee (e.g., ε, δ values) for the generated synthetic data, demonstrating adherence to

differential privacy principles.

NFR4: Data Fidelity: The generated synthetic data shall exhibit high statistical fidelity to the real data, as measured by metrics such as Maximum Mean Discrepancy (MMD)

and preservation of key statistical properties (e.g., mean, variance, correlations, distributions).
NFR5: Robustness: The system shall be robust to variations in input data quality and quantity, producing stable and reliable synthetic data outputs.

NFR6: Security: The system shall implement appropriate security measures to protect the underlying real data used for training and the synthetic data generated.

NFR7: Maintainability: The codebase shall be well-documented, modular, and easy to maintain and extend for future enhancements.

NFR8: Usability (for developers/researchers): The system shall provide clear APIs or interfaces for other Manō components and researchers to interact with for data generation and simulation.

4.3 Expected Test Cases

Expected test cases will validate the functional and non-functional requirements of the system.

TC1: Synthetic Tabular Data Generation:
Input: Request to generate 100 synthetic survey responses conditioned on 'moderate anxiety' and 'age 30-35'.
Expected Output: 100 synthetic tabular records with statistical properties (e.g., DASS scores, demographics) consistent with the specified conditions and similar to real data, and a quantifiable privacy budget.

TC2: Synthetic Time-Series Data Generation:
Input: Request to generate 50 synthetic wearable data sequences (e.g., heart rate variability) for a 'stressed' user profile.
Expected Output: 50 synthetic time-series sequences exhibiting temporal dynamics and statistical properties consistent with stress, and a quantifiable privacy budget.

TC3: Synthetic Text Data Generation:

Input: Request to generate 20 synthetic emotional text entries reflecting 'depressive mood'.

Expected Output: 20 synthetic text entries with linguistic and emotional nuances consistent with depressive mood, and a quantifiable privacy budget.

TC4: Intervention Simulation (Mindfulness):

Input: Generate synthetic data for a user with 'high anxiety', then simulate a mindfulness intervention'.

Expected Output: Post-intervention synthetic data showing a reduction in anxiety-related metrics (e.g., lower DASS anxiety scores, normalized heart rate variability) compared to pre-intervention data, with privacy guarantees.

TC5: Privacy Budget Verification:

Input: Train the GAN with a specified privacy budget (e.g., $\varepsilon=1$, $\delta=10^{-5}$).

Expected Output: The system reports the achieved privacy budget (ε, δ) for the training run, confirming it is within acceptable limits.

TC6: Data Utility Assessment (ML Model Performance):

Input: Train a stress prediction model using the generated synthetic data.

Expected Output: The stress prediction model trained on synthetic data achieves comparable performance (e.g., accuracy, F1-score) to a model trained on real data (within a defined margin), demonstrating high utility.

TC7: Feedback Loop Effectiveness:

Input: Provide simulated real-world chatbot interaction data indicating a discrepancy in synthetic data realism.

Expected Output: The GAN model is retrained or fine-tuned, and subsequent synthetic data generation shows improved realism based on the feedback, with a measurable improvement in fidelity metrics.

5. PROJECT MANAGEMENT

5.1. Gantt Chart

Functional requirements define what the system must do to achieve its objectives.

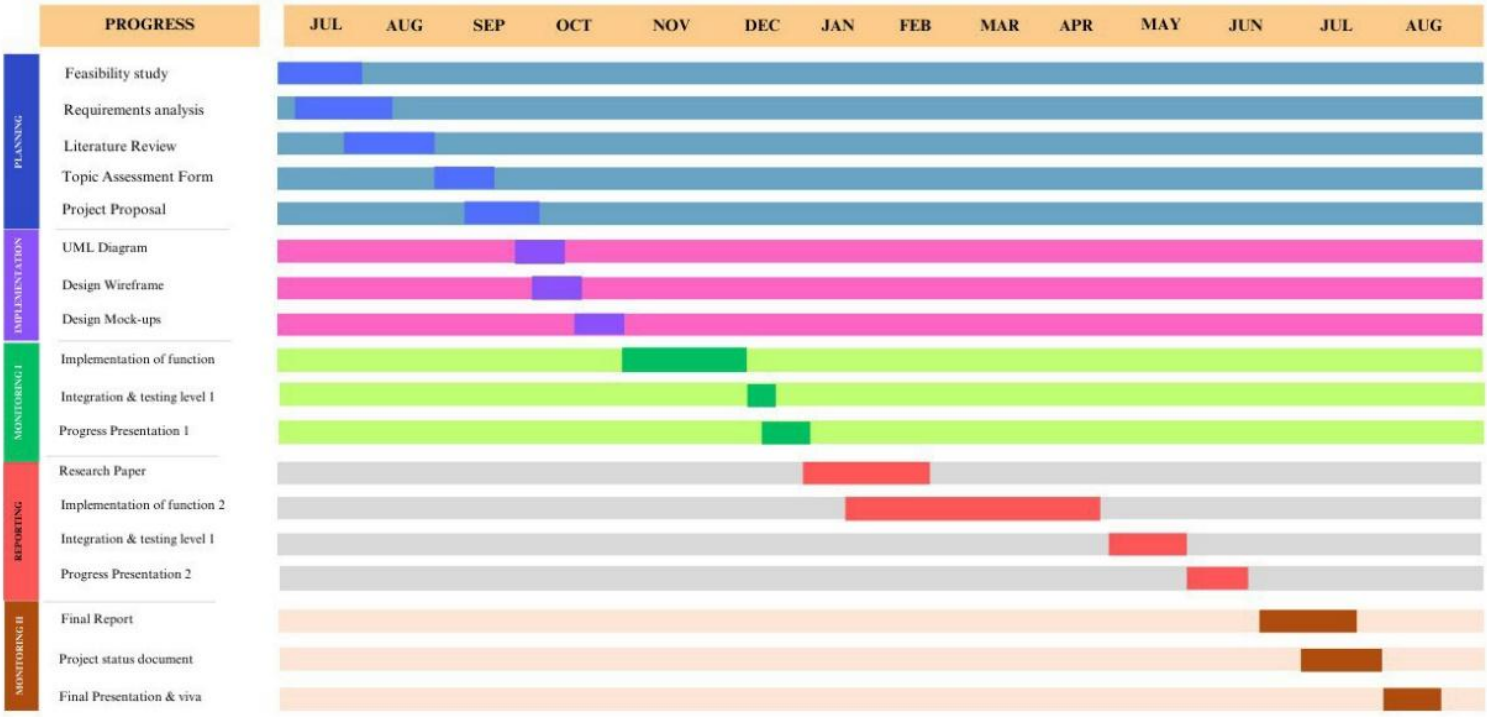

Figure 5.1.1: Gantt chart

6. COMMERCIALIZATION

6.1. Target Audience & User Benefits

The integrated "Manō" framework has significant business potential, offering profound and achievable benefits to a diverse set of stakeholders by addressing their most pressing challenges.

Healthcare Systems & Corporate Wellness Programs:

Benefit: Deploy a holistic mental health support system for employees or patients. Use the prediction system (Component 2) for early risk identification, offer proactive support with the empathetic chatbot (Component 3), and foster a supportive environment through resilience clustering (Component 4). This leads to reduced absenteeism, increased productivity, and lower healthcare costs.

Academic & Clinical Researchers:

Benefit: Gain access to a powerful research platform. Utilize the synthetic data generator (Component 1) to overcome data access barriers, enabling studies that were previously impossible. Use the platform to test hypotheses about risk factors, intervention efficacy, and community dynamics.

Pharmaceutical & Digital Therapeutics (DTx) Companies:

Benefit: Accelerate R&D cycles and de-risk clinical trials. The synthetic data generator allows for the in-silico simulation of new drug or therapy efficacy on diverse, virtual patient cohorts, drastically reducing the time and cost of early-stage trials. The chatbot and clustering components can also serve as platforms for deploying and testing new digital interventions.

Technology Companies & App Developers:

Benefit: License individual components or the entire framework via APIs to enhance their own applications. For example, a general wellness app could integrate the stress prediction model or the empathetic chatbot to add sophisticated mental health features to their product.

6.2. Market Analysis

The global digital mental health market is valued at over \$400 billion and is projected to grow significantly, driven by increased awareness and the demand for accessible, scalable solutions. The "Manō" framework is uniquely positioned to address several key market trends: the shift towards preventative and proactive care, the need for privacy-preserving AI, and the demand for personalized and holistic wellness solutions. By offering an integrated suite of tools, "Manō" can capture a significant share of this burgeoning market.

6.3. Competitive Advantage

"Manō"'s primary competitive advantage is its holistic, four-component integration. While competitors might offer a standalone chatbot or a predictive analytics tool, no current market solution provides an end-to-end framework that:

Generates safe, privacy-preserving synthetic data.
Predict future risk with temporal attention.
Engages users with an empathetic, AI-driven conversational agent.
Connects users into dynamic, supportive peer communities.

This synergistic ecosystem-where each component enhances the others-creates a powerful network effect and a solution that is far more comprehensive and effective than the sum of its parts.

6.4. Business Model

A tiered Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS) model is the most viable route to commercialization, offering flexibility for different customer segments:

Researcher Tier (Low-Cost/Freemium): Provides access to the synthetic data generation module to foster academic research and drive adoption.

Professional Tier (Subscription-based SaaS): Aimed at smaller clinics or companies, offering access to the chatbot and prediction tools for their employees or clients.

Enterprise Tier (PaaS with API Access): Designed for large healthcare systems or corporations. This tier provides full API access to all four components, allowing them to integrate "Manō"'s capabilities into their own platforms, alongside dedicated support and custom model training.

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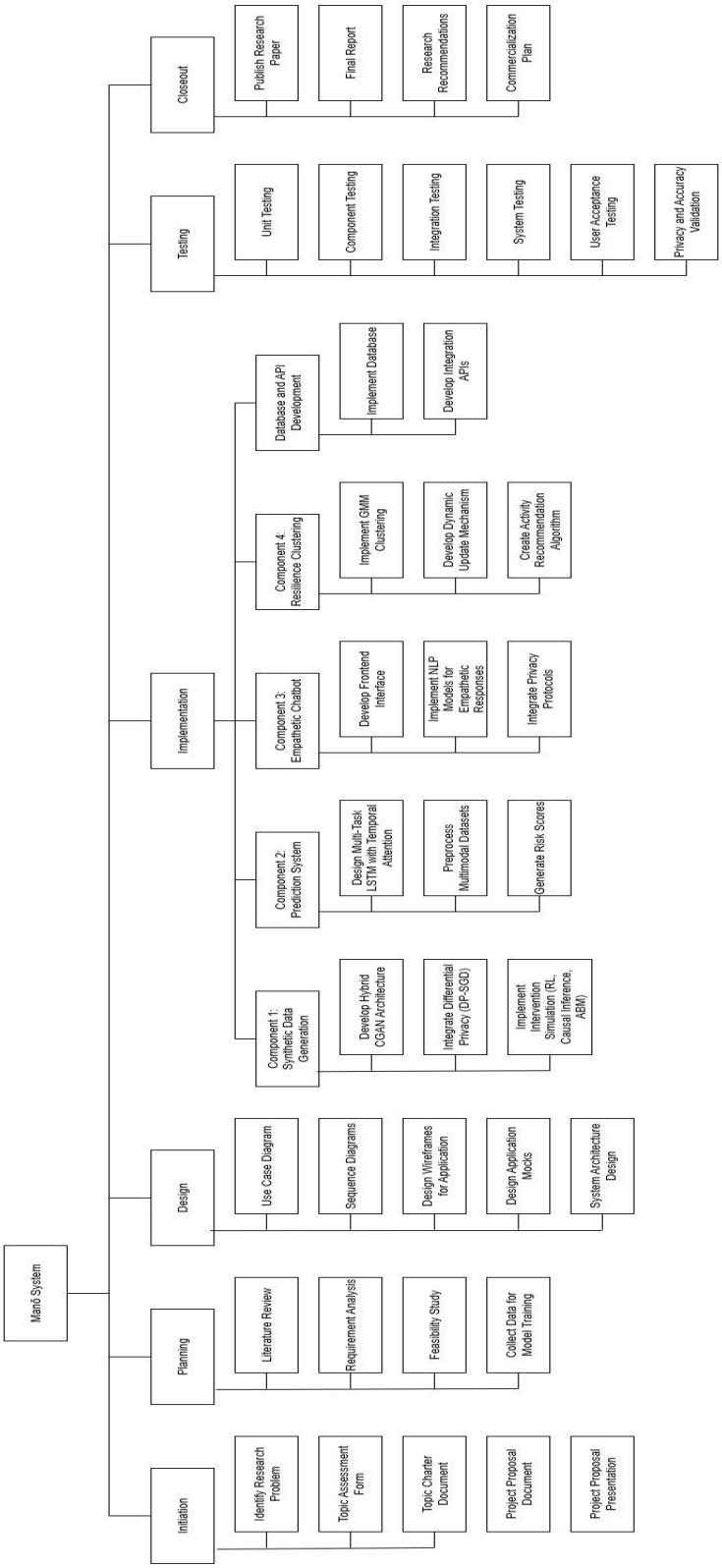
Appendix - A: Work Break Down Chart



Stress, Depression, Anxiety and Cognitive Risk Prediction System (LSTM)

Project Id: 25-26J-163

Project Proposal Report

Keerthi K.K.D.D

B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

Stress, Depression, Anxiety and Cognitive Risk Prediction System (LSTM)

Project Id: 25-26J-163

Project Proposal Report

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Dr. Sanvitha Kasthuriarachchi

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Department of Information Technology

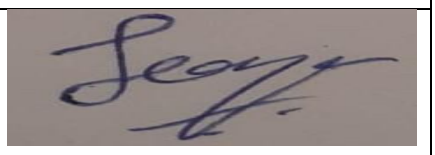
Sri Lanka Institute of Information Technology

Sri Lanka

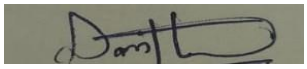
August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Keerthi KKDD	IT22327208	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



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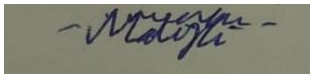

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Signature of the co-supervisor: Date:

ABSTRACT

Mental health disorders constitute a substantial global health burden, affecting approximately 970 million individuals worldwide and imposing economic costs exceeding $2.5 trillion annually. Contemporary predictive frameworks for mental health assessment demonstrate significant limitations in processing heterogeneous multimodal data streams across temporal dimensions, thereby constraining predictive accuracy and individualized intervention capabilities. This research presents a novel Stress, Depression, Anxiety and Cognitive Risk Prediction System leveraging Long Short-Term Memory (LSTM) neural networks augmented with sophisticated temporal attention mechanisms for enhanced mental health risk stratification. The primary research objective centers on developing an integrated predictive framework capable of accurately forecasting individual stress, depression and anxiety trajectories and cognitive performance deterioration through comprehensive multimodal data analysis. The system processes diverse data modalities including structured survey responses, continuous wearable sensor measurements, circadian sleep pattern metrics, and privacy-preserving synthetic datasets generated through Component 1's generative adversarial network architecture. The methodological innovation incorporates a temporal attention mechanism embedded within a multi-task LSTM framework, enabling dynamic prioritization of temporally relevant data points while maintaining historical context awareness for improved prediction granularity. The proposed architecture employs advanced preprocessing techniques for multimodal data fusion, implements a dual-objective LSTM model for simultaneous stress level prediction and cognitive risk assessment, and generates interpretable risk scores for seamless integration with complementary system components. The temporal attention mechanism facilitates adaptive weighting of recent observations while preserving long-term behavioral patterns, addressing the fundamental challenge of temporal relevance in mental health prediction models.Expected research outcomes include statistically significant improvements in predictive accuracy for stress level forecasting and cognitive performance risk assessment compared to conventional approaches, enabling proactive intervention strategies within personalized mental health support ecosystems. This component represents a substantial methodological advancement in predictive mental health analytics, contributing to the development of more accessible, accurate, and ethically sound mental health support systems for diverse user populations across varied sociodemographic contexts.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
BIBM	Bioinformatics and Biomedicine
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
DTx	Digital Therapeutics
eHealth	Electronic Health
EMNLP	Empirical Methods in Natural Language Processing
IEEE	Institute of Electrical and Electronics Engineers
LSTM	Long Short-Term Memory
mHealth	Mobile Health
PaaS	Platform-as-a-Service
PHQ-9	Patient Health Questionnaire-9
PMC	PubMed Central
R&D	Research and Development
RETAIN	Reverse Time Attention Mechanism
RNN	Recurrent Neural Network
SaaS	Software-as-a-Service
WHO	World Health Organization
X-A-BiLSTM	eXtended Attention Bidirectional Long Short-Term Memory

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1. INTRODUCTION

Mental health represents one of the most pressing global health challenges of the 21st century. The World Health Organization indicates that mental disorders contribute to one in six years lived with disability, with severe conditions reducing life expectancy by 10 to 20 years [1]. Despite the substantial economic burden exceeding $2. Trillion annually, governments allocate only approximately 2% of health budgets to mental health care [2]. The emergence of digital health technologies presents unprecedented opportunities to bridge the accessibility gap in mental health support. However, existing solutions face significant limitations in providing personalized, accurate, and timely interventions. Traditional approaches often fail to leverage the rich, multimodal data streams now available through wearable devices, smartphone sensors, and digital interactions [3]. Recent advances in machine learning, particularly in time-series analysis and attention mechanisms, offer promising avenues for developing more sophisticated predictive models. Long Short-Term Memory (LSTM) networks have demonstrated effectiveness in handling temporal sequences, while attention mechanisms have revolutionized how models process and prioritize information [4]. The integration of these technologies within the mental health domain represents critical research opportunity.

This research focuses on developing Component 2 of the Manō framework, a comprehensive mental health support system. The component aims to create an advanced LSTM-based prediction system capable of accurately forecasting stress levels and cognitive performance risks using diverse multimodal data sources. The system will serve as a critical foundation for proactive mental health interventions within the broader Manō ecosystem.

1.1 Background & Literature Survey

1.1.1 Mental health prediction and machine learning

The application of machine learning techniques in mental health prediction has gained significant traction in recent years. Traditional approaches relied heavily on clinical

assessments and self-reported measures, which often suffer from subjectivity and temporal limitations [5]. The advent of digital biomarkers and continuous monitoring through wearable devices has opened new possibilities for objective, real-time mental health assessment [6]. Several studies have demonstrated the potential of machine learning models in predicting mental health outcomes. Sau and Bhakta (2017) explored the use of various machine learning algorithms for depression detection, highlighting the importance of feature selection and model interpretability [7]. Similarly, Thorstad and Wolff (2019) conducted a comprehensive review of machine learning applications in mental health, emphasizing the need for robust validation methodologies [8].

1.1.2 LSTM networks in time-series health prediction

LSTM networks have emerged as a powerful tool for analyzing sequential health data. Their ability to capture long-term dependencies while mitigating the vanishing gradient problem makes them particularly suitable for health-related time series analysis [9]. Choi et al. (2016) demonstrated the effectiveness of LSTM models in predicting healthcare outcomes using electronic health records, achieving superior performance compared to traditional statistical methods [10]. In the context of mental health, several researchers have successfully employed LSTM architectures. Mehta et al. (2019) developed an LSTM-based model for depression detection using social media data, achieving promising results in early detection scenarios [11]. However, most existing approaches focus on single-modal data sources, limiting their potential for comprehensive health assessment.

1.1.3 Attention Mechanisms in Healthcare

Attention mechanisms have revolutionized the field of machine learning by enabling models to focus selectively on relevant information. In healthcare applications, attention mechanisms have proven particularly valuable for interpreting complex, high-dimensional data [12]. Choi et al. (2017) introduced RETAIN, an interpretable predictive model that uses attention mechanisms to identify relevant medical codes and time steps in patient history [13]. The application of temporal attention in mental

health prediction remains relatively unexplored. Most existing studies focus on spatial attention for medical image analysis or simple attention mechanisms for text processing. The integration of sophisticated temporal attention mechanisms within LSTM architectures for multimodal mental health data represents a significant research opportunity.

1.1.4 Multimodal data integration in mental health

The integration of diverse data modalities has shown promise in improving prediction accuracy and providing more comprehensive health insights. Maxhuni et al. (2016) demonstrated the benefits of combining smartphone sensor data with self-reported measures for stress detection [14]. Similarly, Garcia-Ceja et al. (2018) explored the fusion of multiple sensor modalities for depression detection, achieving improved performance compared to single-modal approaches [15]. However, challenges remain in effectively combining heterogeneous data sources with different temporal resolutions and characteristics. The development of robust fusion techniques that can handle missing data, varying sampling rates, and different data types represent an ongoing research challenge.

1.2 Research Gap

Research Paper	Objective	Data Modalities	Attention Mechanism	Risk Score Output
A. P. Bobade *et* al. [16]	Binary stress classification	Physiological signals	No	No
Q. Cong *et* al. [17]	Depression detection from social media	Text data from social media posts	Word-level attention	No
L. Xia *et* al. [18]	EEG-based stress detection	EEG signals only	Hierarchical	No
A. Pandey *et* al. [19]	Text-based mental	Text data only	No	No

	health assessment			
Proposed System	Multi-task stress and cognitive risk prediction with personalized interventions	Surveys, wearables, sleep metrics, synthetic data, behavioral data	Temporal attention	Yes

Table 1.2.1: Characteristics of existing research literature and proposed system

Despite significant progress in applying machine learning to mental health prediction, several critical gaps remain:

Limited Temporal Context Integration: Existing LSTM-based approaches in mental health primarily focus on sequential processing without sophisticated attention mechanisms to prioritize temporally relevant information. Most models treat all historical data points equally, failing to capture the varying importance of recent versus distant observations in mental health assessment. Insufficient Multimodal Integration: While individual studies have explored specific data modalities (wearables, surveys, behavioral data), there is a lack of comprehensive frameworks that effectively integrate diverse multimodal streams with varying temporal characteristics and sampling frequencies.

Lack of Personalized Risk Assessment: Current predictive models often adopt one-size-fits-all approaches, failing to account for individual differences in stress manifestation and cognitive performance patterns. The development of personalized models that can adapt to diverse user profiles remains limited.

Interpretability Challenges: Most existing deep learning approaches for mental health prediction operate as black boxes, providing limited insights into the factors driving

predictions. This lack of interpretability hinders clinical adoption and trust in automated systems.

Real-time Processing Limitations: Many current approaches are designed for offline analysis rather than real-time prediction, limiting their utility for proactive intervention systems that require immediate risk assessment.

This research addresses these gaps by developing a novel multi-task LSTM architecture with temporal attention mechanisms specifically designed for real-time, personalized mental health risk prediction using diverse multimodal data sources.

1.3 Research Problem

How can we develop an accurate, real-time, and personalized system for predicting stress levels and cognitive performance risks that effectively integrates multimodal temporal data while providing interpretable risk assessments for proactive mental health interventions

Predicting mental health outcomes from multimodal temporal data poses several challenges that current models struggle to address effectively. Traditional LSTM architectures capture sequential dependencies but often fail to highlight the most critical temporal patterns, limiting prediction accuracy. Existing approaches also lack personalization, as they do not adapt to individual user profiles or evolving behavioral patterns, which is crucial for accurate mental health risk assessment. Additionally, many models produce opaque risk scores that hinder interpretability and reduce their clinical utility. Finally, real-time performance remains a challenge, as models must efficiently process complex multimodal temporal sequences without sacrificing accuracy or scalability.

2. OBJECTIVES

2.1 Main Objectives

To accurately predict individual stress, depression, anxiety levels and cognitive performance risks using multimodal data, enabling proactive and personalized interventions within the Manō mental health framework.

2.2 Specific Objectives

A multi-task LSTM architecture will be designed and implemented to enable simultaneous prediction of stress, depression, anxiety and cognitive risk.

To enhance predictive performance, a temporal attention mechanism will be integrated within the LSTM to prioritize recent and relevant data points,

Interpretable risk scores will be generated ensuring integration with other system components such as chatbots and clustering modules

3. METHODOLOGY

3.1 Overall System Description

The Stress, Depression, Anxiety and Cognitive Risk Prediction System is designed as Component 2 of the Manō framework, serving as the predictive analytics engine that processes multimodal temporal data to generate real-time risk assessments. The system receives input from multiple sources and provides risk scores to other components for personalized interventions

3.2 System Diagram

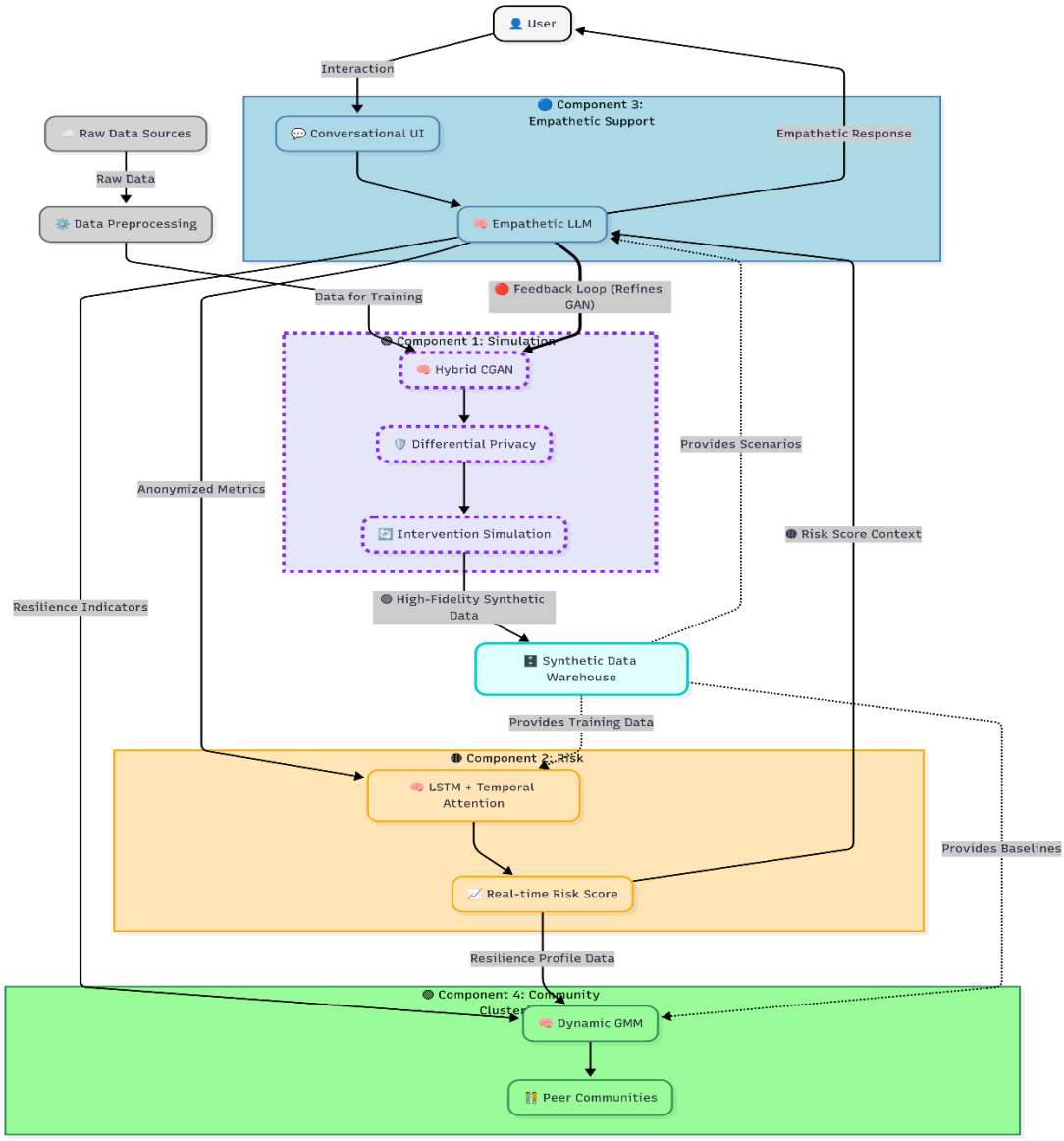

Figure 3.2.1: System overview diagram

3.3 Individual System Diagram

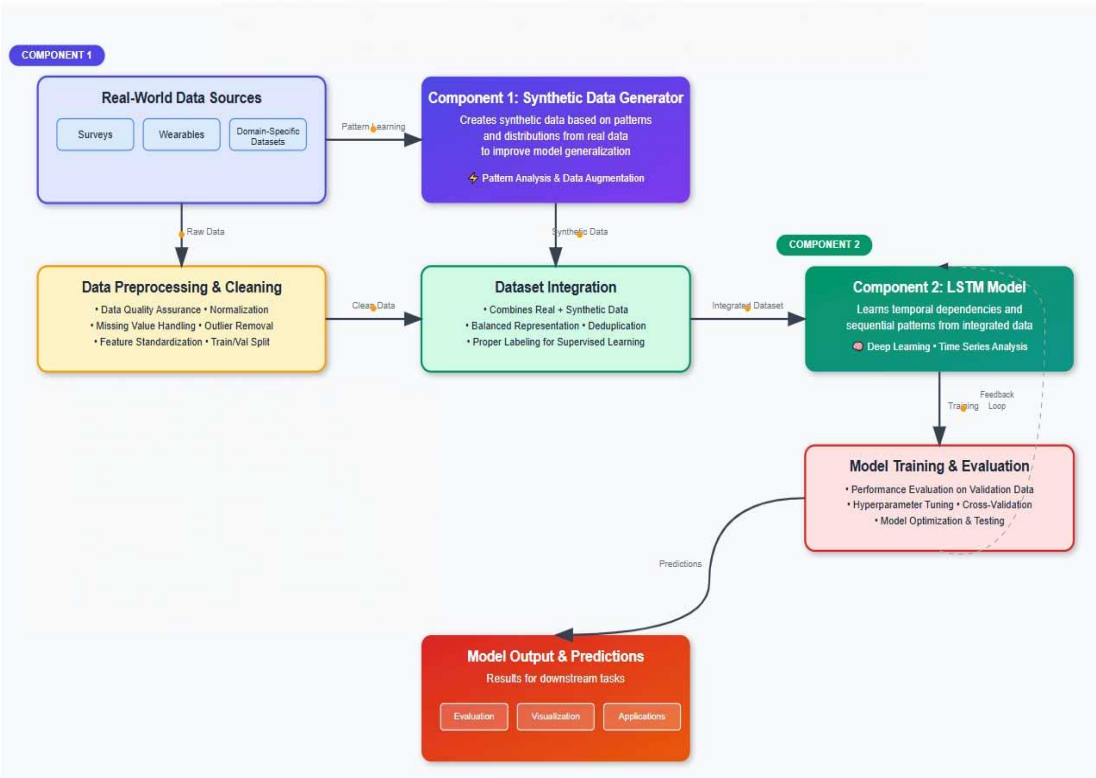

Figure 3.3.1: Individual system diagram

3.4 Explanation of the System Diagram

The diagram illustrates the system consisting of two main components along with the data processing pipeline. Component 1, the Synthetic Data Generator, creates synthetic data based on patterns or distributions derived from real data, providing additional training samples to improve model generalization. Real-world data sources include surveys, wearables, and other domain-specific datasets, which undergo preprocessing to ensure data quality, normalization, and handling of missing values. During data preprocessing and cleaning, the data is standardized, outliers are removed, features are normalized, and the dataset is split into training and validation sets. The dataset integration stage maintaining proper labeling for supervised learning. LSTM model, is trained in the integrated dataset, learning temporal dependencies and sequential patterns, and its performance is evaluated on validation and test data. Finally, the

model output and predictions provide results for downstream tasks and can be utilized for evaluation, visualization, or real-world applications.

4. PROJECT REQUIREMENTS

This section outlines the key requirements for the Privacy-Preserving Mental Health Simulation System, categorized into functional, non-functional, and expected test cases.

4.1 Functional Requirements

Functional requirements define what the system must do to achieve its objectives.

Stress, Depression and Anxiety Level Prediction: The system shall predict stress levels on a continuous scale from 0-1 The system shall categorize stress predictions into five risk levels The system shall provide confidence intervals for stress predictions

Cognitive Risk Assessment: The system shall predict cognitive performance risks including attention deficits and memory impairment The system shall forecast potential cognitive decline based on temporal patterns The system shall generate early warning indicators for cognitive risk factors

Temporal Pattern Analysis: The system shall analyze temporal dependencies in user data over multiple time horizons. The system shall identify recurring patterns and trends in mental health indicators The system shall adapt predictions based on seasonal and cyclical variations.

Personalized Risk Scoring: The system shall generate personalized risk scores adapted to individual user profiles

Integration Interface: The system shall provide RESTful API endpoints for real-time risk score requests. The system shall support batch processing for historical data

analysis. The system shall maintain integration protocols with Components 1, 3, and 4 of the Manō framework

4.2 Non-Functional Requirements

Non-functional requirements specify criteria that can be used to judge the operation of a system, rather than specific behaviors.

Performance & Accuracy: The system must provide accurate and timely predictions of stress and cognitive risks.

Scalability: The architecture should handle increasing numbers of users and support deployment on cloud platforms.

Reliability & Availability: The system should operate consistently and handle missing data gracefully.

Interoperability: Predictions must integrate seamlessly with other Manō components via standard APIs.

Security & Privacy: All data must be handled securely, following relevant privacy regulations.

Usability: Outputs should be interpretable and easily understood by technical and non-technical users.

Maintainability: The system should support modular retraining and include documentation for reproducibility and future improvements.

4.3 Expected Test Cases

Expected test cases will validate the functional and non-functional requirements of the system.

Test Case 1: Verify that real data is correctly loaded and preprocessed (e.g., normalization, missing value handling).
Test Case 2: Verify handling of edge cases like empty datasets, corrupted files, or unexpected data types.

Test Case 3: Verify that the LSTM model can start training with the combined dataset without errors.

Test Case 4: Test that training progresses correctly over epochs and logs loss/accuracy metrics.

Test Case 5: Check the model's ability to handle hyperparameter changes (e.g., number of layers, hidden units).

Test Case 6: Verify early stopping, checkpoint saving, and model retraining when new data is added.

Test Case 7: Evaluate prediction accuracy on a validation set using only real data.

Test Case 8: Evaluate performance on a validation set using synthetic data.

Test Case 9: Test the model for overfitting or underfitting with small and large datasets.

5. PROJECT MANAGEMENT

5.1. Gantt Chart

Functional requirements define what the system must do to achieve its objectives.

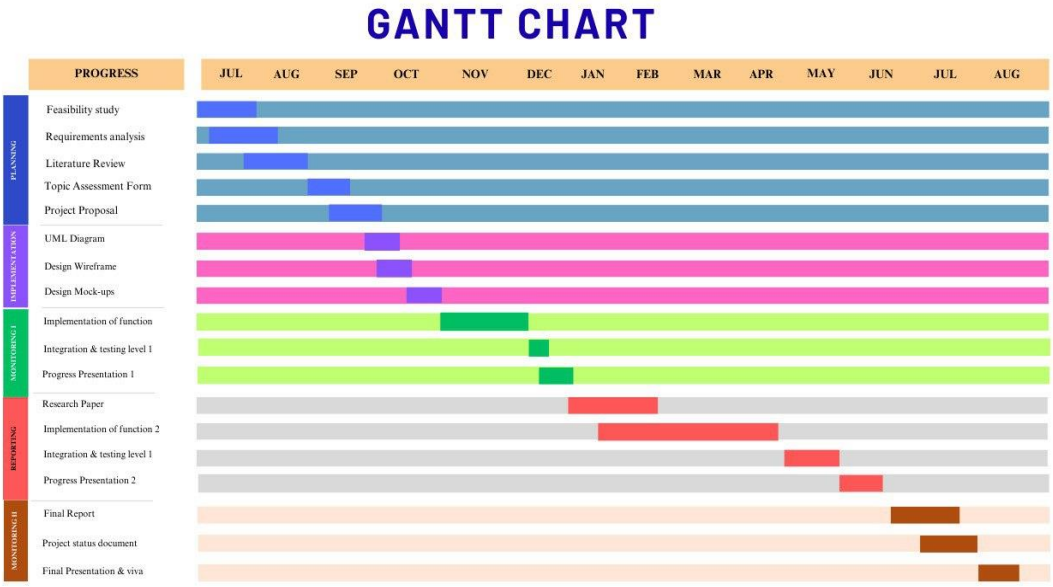

Figure 5.1.1: Gantt chart

6. COMMERCIALIZATION

6.1. Target Audience & User Benefits

The integrated "Manō" framework has significant business potential, offering profound and achievable benefits to a diverse set of stakeholders by addressing their most pressing challenges.

Healthcare Systems & Corporate Wellness Programs:

Benefit: Deploy a holistic mental health support system for employees or patients. Use the prediction system (Component 2) for early risk identification, offer proactive support with the empathetic chatbot (Component 3), and foster a supportive environment through resilience clustering (Component 4). This leads to reduced absenteeism, increased productivity, and lower healthcare costs.

Academic & Clinical Researchers:

Benefit: Gain access to a powerful research platform. Utilize the synthetic data generator (Component 1) to overcome data access barriers, enabling studies that were previously impossible. Use the platform to test hypotheses about risk factors, intervention efficacy, and community dynamics.

Pharmaceutical & Digital Therapeutics (DTx) Companies:

Benefit: Accelerate R&D cycles and de-risk clinical trials. The synthetic data generator allows for the in-silico simulation of new drug or therapy efficacy on diverse, virtual patient cohorts, drastically reducing the time and cost of early-stage trials. The chatbot and clustering components can also serve as platforms for deploying and testing new digital interventions.

Technology Companies & App Developers:

Benefit: License individual components or the entire framework via APIs to enhance their own applications. For example, a general wellness app could integrate the stress prediction model or the empathetic chatbot to add sophisticated mental health features to their product.

6.2. Market Analysis

The global digital mental health market is valued at over \$400 billion and is projected to grow significantly, driven by increased awareness and the demand for accessible, scalable solutions. The "Manō" framework is uniquely positioned to address several key market trends: the shift towards preventative and proactive care, the need for privacy-preserving AI, and the demand for personalized and holistic wellness solutions. By offering an integrated suite of tools, "Manō" can capture a significant share of this burgeoning market.

6.3. Competitive Advantage

"Manō"'s primary competitive advantage is its holistic, four-component integration. While competitors might offer a standalone chatbot or a predictive analytics tool, no current market solution provides an end-to-end framework that:

Generates safe, privacy-preserving synthetic data.
Predict future risk with temporal attention.
Engages users with an empathetic, AI-driven conversational agent.
Connects users into dynamic, supportive peer communities.

This synergistic ecosystem-where each component enhances the others-creates a powerful network effect and a solution that is far more comprehensive and effective than the sum of its parts.

6.4. Business Model

A tiered Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS) model is the most viable route to commercialization, offering flexibility for different customer segments:

Researcher Tier (Low-Cost/Freemium): Provides access to the synthetic data generation module to foster academic research and drive adoption.

Professional Tier (Subscription-based SaaS): Aimed at smaller clinics or companies, offering access to the chatbot and prediction tools for their employees or clients.

Enterprise Tier (PaaS with API Access): Designed for large healthcare systems or corporations. This tier provides full API access to all four components, allowing them to integrate "Manō"'s capabilities into their own platforms, alongside dedicated support and custom model training.

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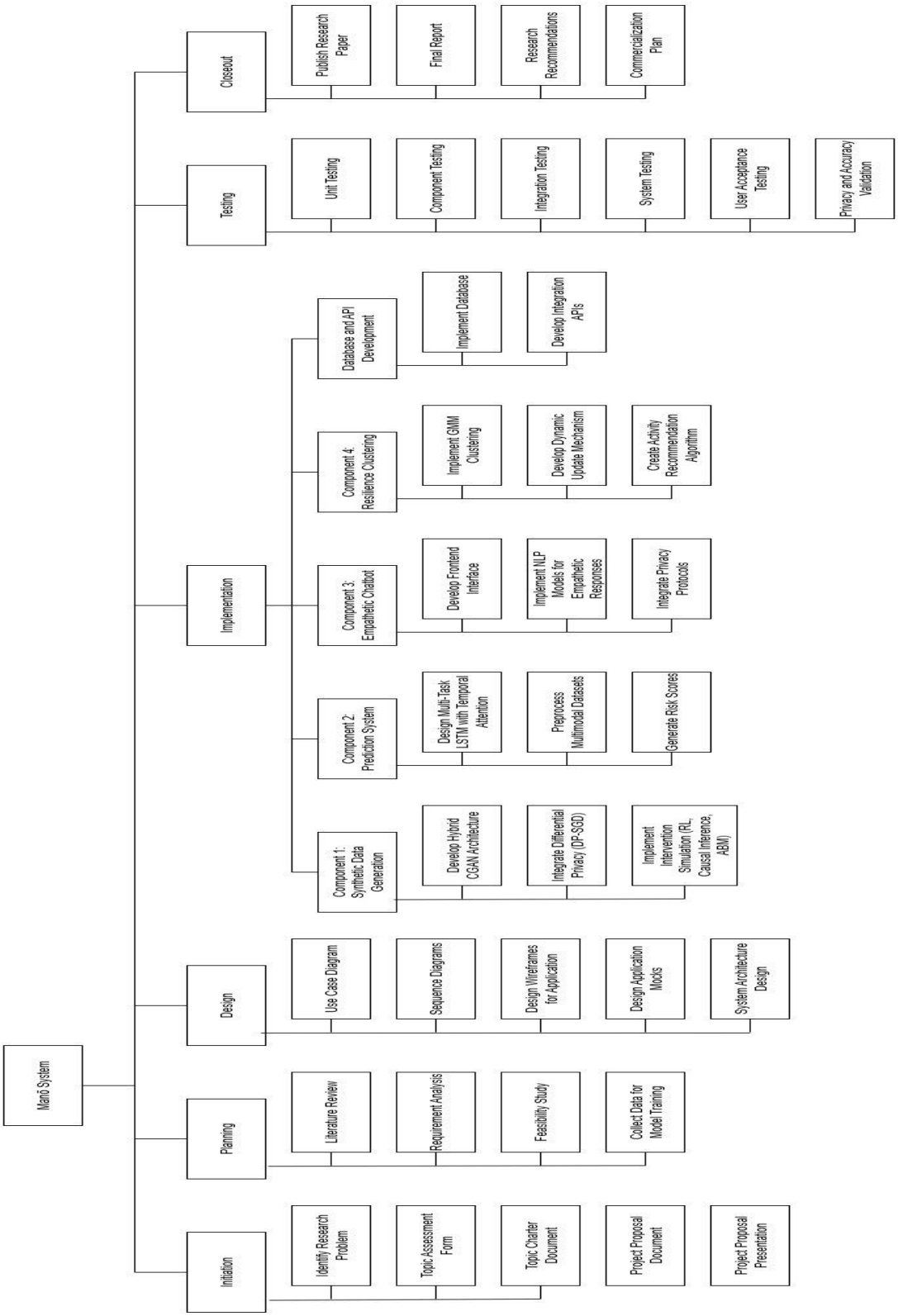
Appendix - A: Work Break Down Chart



Manō : Holistic Framework for Personalized and Community-Driven Cognitive, Emotional, and Resilient Vitality

25 – 26J – 163

Project Proposal Report

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BSc (Hons) Degree in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

Manō : Holistic Framework for Personalized and Community-Driven Cognitive, Emotional, and Resilient Vitality

25 – 26J – 369

Project Proposal Report

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August 2025

Declaration

We hereby confirm that this work is original, and this proposal contains without acknowledgement no material which has been previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief contains no material previously published or written by any other person except where the acknowledgement is expressed in the text.

Student Name	**Student ID**	**Signature**
Wickramasuriya C.I.	IT22890528	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date:

Signature of the co-supervisor: Date:

Abstract

The Empathetic Conversational Support System (Component 3) is a core module of the Manō platform, with the goal to provide STEM professionals with privacy-preserving, personalized mental health support through natural language conversations. This component leverages synthetic data from privacy-preserving generative models for training and simulating empathetic chatbot responses so that no real user data is exposed during development or deployment. The platform uses advanced natural language processing (NLP) techniques, including large language models conditioned on mental health contexts, to deliver supportive conversation that addresses stress, burnout, and cognitive risks common in high-stress STEM environments.

The overall design has a frontend interface to receive user input, backed by a conversational AI engine with differential privacy protocols for maintaining ethical standards. This proposal is for the development of an empathetic chatbot that simulates therapy sessions, provides intervention suggestions, and integrates with other parts of Manō for holistic mental health support. The outcome will be toward readily accessible digital mental health services, with both individual well-being and societal implications for privacy-preserving AI applications.

Keywords: Crowdsources mapping, Localization

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1. Introduction

Mental illness among STEM professionals such as stress and burnout is increasing due to stressful working conditions [1]. Manō platform fights back by a privacy-preserving set of components where Component 3 is aimed at empathetic chat support. The component is completed based on Component 1's data of synthetic generation to render secure, non-intrusive chatbot transactions that imitate professional counseling without reaping personal information.

Some of the applicable AI and NLP work in mental health includes use of bots like Woebot for cognitive behavior therapy [2], which can lower anxiety levels. Practices like conditional GANs for data synthesis [from provided design guide] are tuned to be privacy-aware. The talent now includes use of differential privacy with conversational AI, e.g., models like GPT variants targeted at empathy [3]. Previous studies, such as ELIZA-like systems, have been superficial in scope, while our strategy is distinguished by the use of privacy guarantees and simulation data fusion for maximizing response proximity.

This capability will superimpose these with adaptive language that evolves over time based on aggregate, anonymized trends from conversation history, from the older rule-based systems to the newer AI-based empathy.

1.1 Background & Literature survey

The technology of conversational AI for mental wellness has evolved considerably. Early programs like ELIZA (1966) simulated psychotherapists using pattern matching but were superficial [4]. Recent attempts are Wysa and Woebot, which use NLP to deliver CBT-directed interventions, with depression symptom reduction of 20-30% in user trials [2].

Literature also documents the application of privacy-protection techniques such as differential privacy in AI models for protecting health-sensitive information [5]. Conditional variants in GANs are utilized to generate synthetic conversation sets with statistical fidelity [design guide reference]. Empathetic AI studies such as that using transformer models (e.g., BlenderBot) aim to generate responses conditioned on emotional state [6].

Knowledge contribution: This aspect propels scientific progress by the collaboration of synthetic data from privacy-preserving simulations and moment-by-moment empathetic feedback, affecting fund providers via ethical AI innovation and society via inexpensive mental health services.

1.2 Research Gap

Most mental health chatbots that are already out there use real user data to train, which could lead to privacy issues, or they don't work with bigger simulation systems to provide personalised help. Woebot is a general support tool, but it doesn't use differential privacy or synthetic scenario generation for STEM-specific situations [2]. There is GAN-based data synthesis, but it isn't often used for conversational empathy in places where privacy is important [design guide]. The problem is that there isn't a system that uses synthetic profiles that protect privacy to train empathetic responses without exposing real data. This makes it harder for professionals to work in sensitive areas like mental health.

1.3 Research Problem

The principal issue addressed is the lack of privacy-protecting, empathetic conversational interfaces particularly for STEM professionals with mental health issues. For example:
How do AI chatbots provide personalized assistance using simulated data without infringing upon user privacy? What are the implications of simulation-based interventions in the flow of conversations? How do these systems portray, respond to, and narrate users' concerns without compromising ethical aspects?

2. Objectives

Objectives direct what the research is to do, with SMART objectives to evaluate.

2.1 Primary Objectives

To develop an empathetic dialogue support system utilizing privacy-preserving synthetic data to provide personalized mental health assistance for STEM employees, making links among user questions, virtual interventions, and empathetic responses.

2.2 Specific Objectives

To develop a frontend user interface for natural, conversational interaction and synthesized-on profiles from Component 1.

To leverage NLP models trained in mental health scenarios for generating empathetic responses.

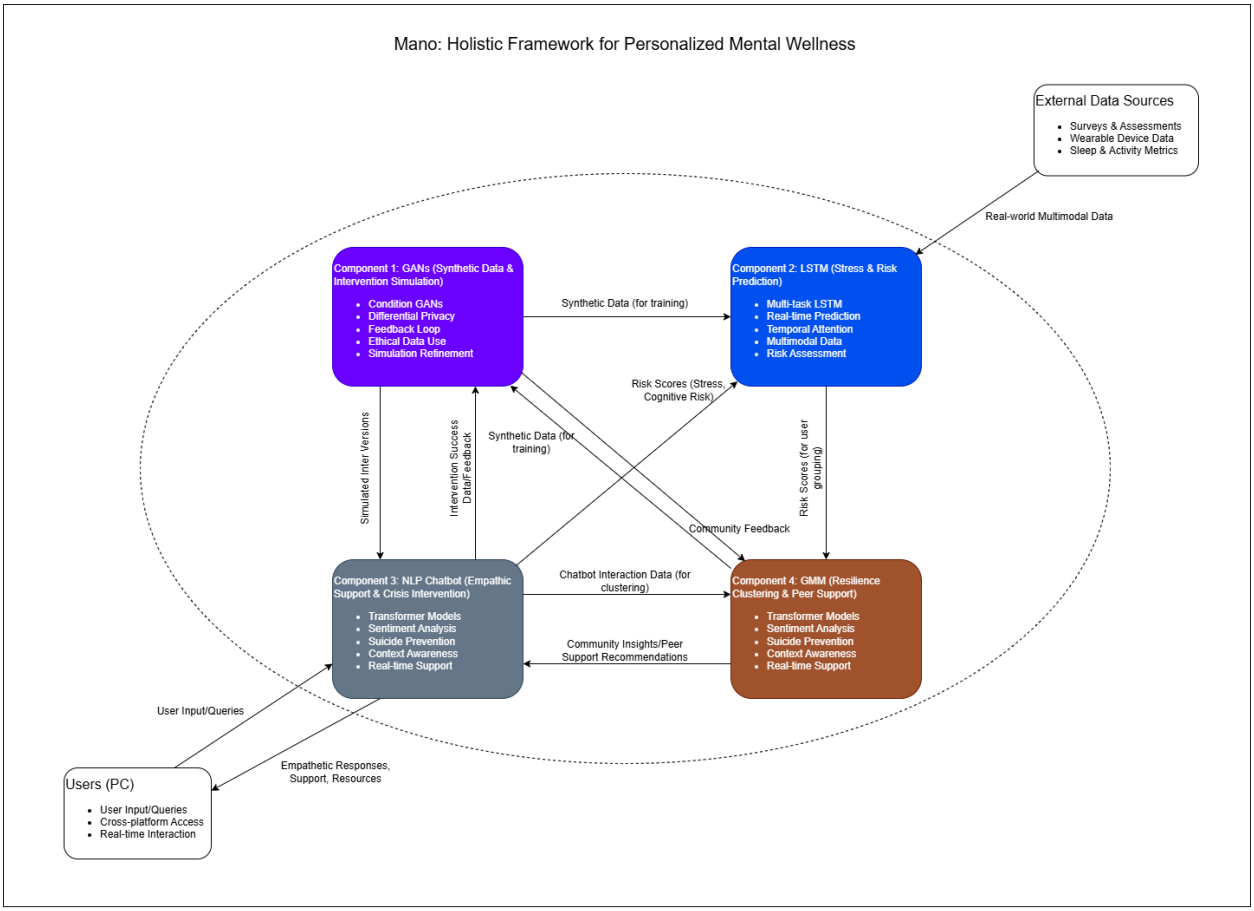
To offer differential privacy in all data handling and response generation.

To quantify the system's effectiveness in emulating therapeutic conversations using user testing and metrics like response relevance.

3. Methodology

The methodology describes the scientific process of developing the Empathetic Conversational Support System (Component 3) of the Manō platform. The component aims to design an AI-powered chatbot delivering empathetic, privacy-preserving mental health care tailored to STEM professionals. The development process integrates frontend user interfaces and backend NLP models with synthetic data from Component 1 to facilitate ethical data processing and differential privacy. It is an iterative process, leveraging agile techniques to facilitate development through application of feedback from test, as promoted in guidelines for research projects.

3.1 System Diagram



3.2 Overall System

The system is built as a modular single-page application (SPA) with a React-based frontend for real-time conversational interaction and a Python backend with transformer models (e.g., a fine-tuned version of GPT-3 or other large language models) for response generation. The chatbot simulates empathetic conversations by conditioning responses on synthetic mental health profiles obtained from Component 1's Privacy-Preserving Mental Health Simulation System. This ensures that no real user data is used in training or inference, with strict privacy guarantees through the use of differential privacy mechanisms, where noise is added to model outputs to prevent reverse-engineering of inputs [design guide, adapted from Component.

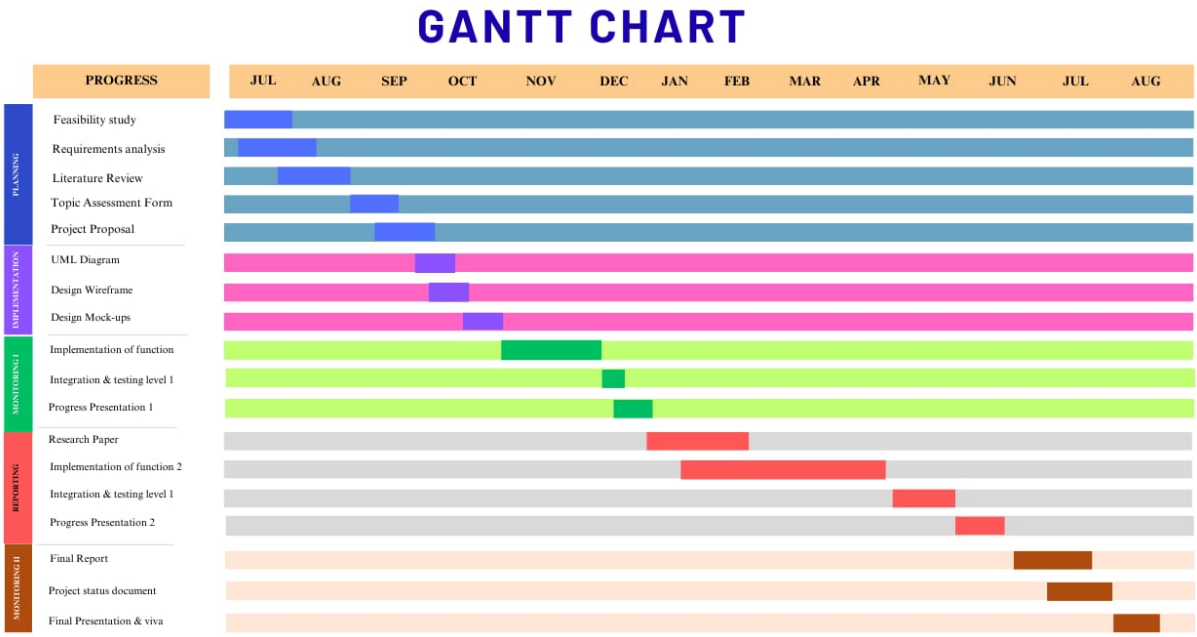
The chief workflow is for users to interact via chat interface, with inputs interpreted for emotional context, conditioned against synthetic scenarios (e.g., stress levels in STEM environments), and used to provide supportive responses. Integration with the other Manō components involves receiving synthetic conversation scenarios from Component 1 for training, sending anonymized interaction patterns to Component 2 for stress prediction refinement, and sharing resilience insights with Component 4 for peer matching.

3.3 Implementation Step

To facilitate methodical development and integration, the Empathetic Conversational Support System (Component 3) rollout is divided into five essential steps:

- Conditions Collecting and Design: Determine system requirements from Component 1's synthetic data specifications, including UI/UX and NLP model requirements. Make chat interface wireframes.
- Development of Frontends: Create a React single-page application with a real-time chat user interface, perform utilizing WebSocket to offer accessibility features and a smooth user experience (WCAG 2.1 compliant).
- Implementation of Backend NLP: Use a transformer-based LLM (such as fine-tuned GPT) with differential privacy to deploy the NLP engine. Use Component 1's synthetic mental health examples to train the model.
- System Testing and Integration: RESTful APIs are used to make it easier to move data from the frontend to the backend. usability testing on simulated users, integration testing (Component 1 data pipeline), and unit testing (NLP accuracy).
- Deployment and Evaluation: Set up the system on an AWS or other cloud computing platform, iterate on privacy scans and user satisfaction metrics to evaluate performance before deployment.

3.4 Gannt Chart



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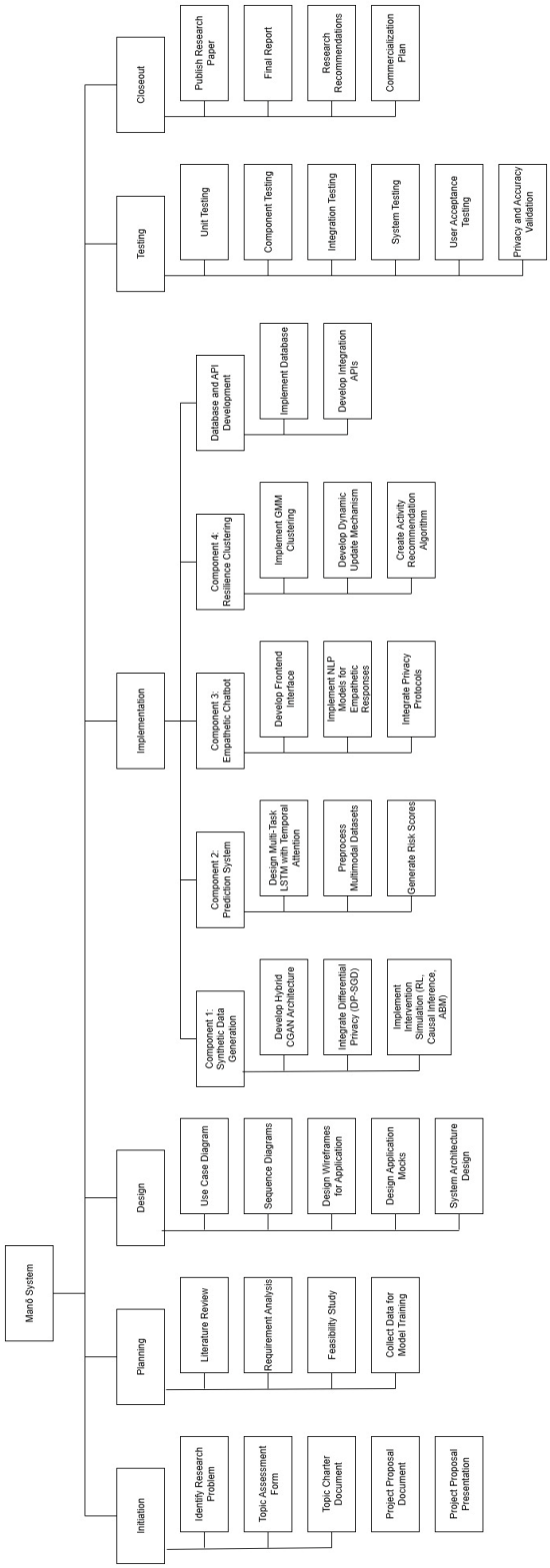

4. Project Requirement

4.1 Functional Requirements

User authentication using anonymous sessions.

Text-based conversational interface with real-time response.

Empathic response generation from synthesized scenarios.

Import of data via integration with Component 1.

Export of conversation summaries (anonymous).

4.2 Non-Functional Requirements

Latency in response < 2 seconds.

99% availability.

Compliance with GDPR and differential privacy regulations.

Scalability to 1000 concurrent users.

Accessible UI (WCAG 2.1 AA).

4.3 Expected Test Cases

Unit tests: NLP response accuracy (e.g., input "I'm stressed" → empathetic response).

Integration tests: Data import from Component 1.

Usability tests: User satisfaction surveys.

Privacy tests: Attempt to reverse-engineer artificial responses (should fail).

5. Commercialization and Entrepreneurship Potential

5.1. Target Audience & User Benefits

The integrated "Manō" framework has significant business potential, offering profound and achievable benefits to a diverse set of stakeholders by addressing their most pressing challenges.

- Healthcare Systems & Corporate Wellness Programs:
    - Benefit: Deploy a holistic mental health support system for employees or patients. Use the prediction system (Component 2) for early risk identification, offer proactive support with the empathetic chatbot (Component 3), and foster a supportive environment through resilience clustering (Component 4). This leads to reduced absenteeism, increased productivity, and lower healthcare costs.
- Academic & Clinical Researchers:
    - Benefit: Gain access to a powerful research platform. Utilize the synthetic data generator (Component 1) to overcome data access barriers, enabling studies that were previously impossible. Use the platform to test hypotheses about risk factors, intervention efficacy, and community dynamics.
- Pharmaceutical & Digital Therapeutics (DTx) Companies:
    - Benefit: Accelerate R&D cycles and de-risk clinical trials. The synthetic data generator allows for the *in-silico* simulation of new drug or therapy efficacy on diverse, virtual patient cohorts, drastically reducing the time and cost of early-stage trials. The chatbot and clustering components can also serve as platforms for deploying and testing new digital interventions.
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6. Reference list

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7. Appendix

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Manō : Holistic Framework for Personalized and Community-Driven Cognitive, Emotional, and Resilient Vitality 25 – 26J – 163 Project Proposal Report Wickramasuriya C.I. BSc (Hons) Degree in Information Technology Specialized in Information Technology Department of Information Technology Sri Lanka Institute of Information Technology Sri Lanka August 2025 Manō : Holistic Framework for Personalized and Community-Driven Cognitive, Emotional, and Resilient Vitality 25 – 26J – 369 Project Proposal Report Wickramasuriya C. I. – IT22890528 Supervisor – Ms. Anjalie Gamage BSc (Hons) Degree in Information Technology Specialized in Information Technology Department of Information Technology Sri Lanka Institute of Information Technology Sri Lanka August 2025 Declaration We hereby confirm that this work is original, and this proposal contains without acknowledgement no material which has been previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief contains no material previously published or written by any other person except where the acknowledgement is expressed in the text. Student Name Student ID Signature Wickramasuriya C.I. IT22890528 The supervisor/s should certify the proposal

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